

TANVIR SIDDIQUE

Computer Science & Engineering • Khulna University of Engineering & Technology

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EDUCATION

Khulna University of Engineering & Technology (KUET) Feb 2026 — Present

B.Sc. in Computer Science & Engineering

Undergraduate coursework in algorithms, data structures, systems programming, and software engineering fundamentals.

Notre Dame College, Dhaka 2023 — 2025

Higher Secondary Certificate (HSC)

Science division coursework with focus on physics, chemistry, and higher mathematics.

Govt Naldanga Bhushan Pilot Secondary School 2018 — 2022

Secondary School Certificate (SSC)

Secondary education in science division with distinction.

HONORS & AWARDS

- **Champion (1st Place)** — ForkedArch Freshers Hackathon 2026 (Forkathon 2026) for CampusShare KUET.
- **46,900+ Automated Bot Edits** — Executed across global Wikimedia projects maintaining taxonomy and category structures.

TECHNICAL SKILLS

Languages	Python, TypeScript / JavaScript, C / C++, SQL (PostgreSQL), Git & GitHub
Frameworks & Tools	FastAPI & REST APIs, React 18 & Vite, VS Code Extension API, Wikimedia Toolforge & Pywikibot, Streamlit & Plotly, TensorFlow / Keras & PyTorch, Numpy & Pandas, Capacitor Mobile Runtime
Ecosystem & Domains	Open Source Coordination, Metadata & Categorization, Technical Operations, Linguistics & Classical Arabic, Community Mentorship

OPEN SOURCE & TECHNICAL EXPERIENCE

Lead of Operations • Project Koushal Sep 2024 — Present

- Direct technical and operational logistics for nationwide community learning initiatives.
- Streamline communication channels, workflow pipelines, and resource distribution.

Team Member & Data Lead • Project Korikath Jan 2024 — Present

- Architected and maintain CampAnalytics on Wikimedia Toolforge for campaign analytics and retention tracking.
- Develop data pipelines analyzing contributor churn, event health scorecards, and cross-wiki metrics.

Contributor & Bot Operator • Wikimedia Foundation Jun 2022 — Present

- Engineer automated Pywikibot pipelines for cross-wiki categorization and structural maintenance.
- Structured ~10,000 articles on Fulani Wikipedia and categorized over 15,800 media files on Wikimedia Commons.

Category Manager • Wiki Loves Folklore Feb 2025 — Apr 2025

- Curated and managed media metadata taxonomy for international photo competition submissions.
- Audited bulk uploads and ensured compliance with Wikimedia Commons categorization standards.

KEY TECHNICAL PROJECTS

CampAnalytics — Event Evaluation & Campaign Analytics [[Code](#) | [Live on Wikimedia](#) Python, Flask, Streamlit, Pandas Toolforge | [Interactive Streamlit Dashboard](#)]

Analytics tool built for Project Korikath to evaluate Wikimedia Commons campaigns (WLM, WLE, WLF, WLA, WLB) and contributor retention. Deployed on Wikimedia Toolforge and Streamlit.

CodeFunc [[Code](#) | [VS Code Marketplace \(v1.0.3\)](#)]

TypeScript, VS Code Extension, Chrome Extension, Gemini API

VS Code and Chrome extension providing an architectural file overview at Line 0. Features fast local AST parsing and contextual code summaries with Gemini, Groq, and Claude.

CampusShare KUET [[Code](#) | [Live Web App](#) | [Download Android APK](#)]

React 18, FastAPI, PostgreSQL, MapLibre GL

1st Place (Champion) at Forkathon 2026. A peer-to-peer equipment borrowing platform for KUET residential halls. Built in 50 hours with FastAPI, React, MapLibre GL radar, OTP/QR handover verification, and an Android app.

AgroScan — Crop Disease Detection & Smart Farm [[Code](#) | [Live AgroScan App](#) Python, TensorFlow, Keras, EfficientNet | [Integrated in Smart Farm Repo](#)]

38-class plant pathology transfer learning model (crop_disease_model_final.keras) trained on leaf photos. Built from scratch with a Streamlit interface (AgroScan) and integrated into the Smart Farm project.

TanvirSdqBot — Wikimedia Automation Engine [[Code](#)]

Python, Pywikibot, MediaWiki API, Pandas

Pywikibot scripts used to structure ~10,000 articles on Fulfulde Wikipedia, automate welcome messages for new editors, and process event data for Wiki Loves campaigns.

Kaggle ML & Data Science Workspace [[Code](#)]

Python, Scikit-Learn, XGBoost, LightGBM

Hands-on repository of Kaggle competition solutions and coursework from the University of Tokyo Matsuo-Iwasawa Lab GCI World 2026 program.

MLP & Language Modeling Lab [[Code](#)]

Python, PyTorch, NLP, Bigram LM

Character-level Bigram Language Model trained on The Wizard of Oz in PyTorch, exploring tokenization, cross-entropy loss, and autoregressive text generation.

Algorithmic Problem Solving [[Code](#)]

C, C++17, STL, Algorithms

Algorithmic problem solving practice in C and modern C++ STL for university coursework and programming fundamentals.

EXTRACURRICULAR LEADERSHIP & SOCIETIES

Director of Leadership & Management • Notre Dame English Club (WERT)

Sep 2024 — Sep 2025

- Led leadership operations and managed event coordination for college-wide literary and cultural programs.
- Mentored junior executive members and coordinated inter-departmental communications.

Member • Hardware Acceleration Club of KUET (HACK)

Jul 2026 — Present

Member • Special Group Interested In Programming Contest (SGIPC)

Jul 2026 — Present

Member • KUET Photographic Society (KUETPS)

Jul 2026 — Present

RELEVANT COURSEWORK & TRAINING

- Machine Learning Specialization (Andrew Ng, Coursera) (completed)
- Introduction to AI & Data Science (GCI World 2026, Matsuo-Iwasawa Lab U-Tokyo) (ongoing)